

Total No. of Questions : 8]

[Total No. of Printed Pages : 4

Roll No

EC-3003-CBGS

B.E. III Semester

Examination, June 2020

Choice Based Grading System (CBGS)

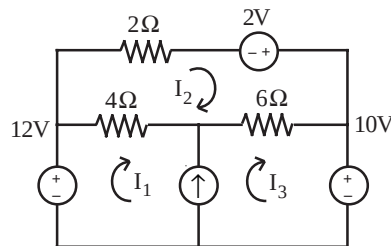
Network Analysis

Time : Three Hours

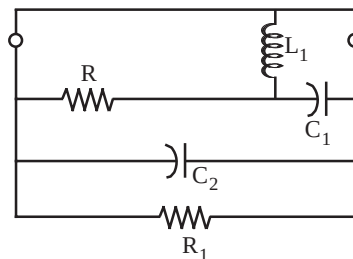
Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) State KCL and KVL and determine the mesh current I_1 , I_2 and I_3 . 7



- b) Define graph of the network and draw the graph for the given circuit. 7

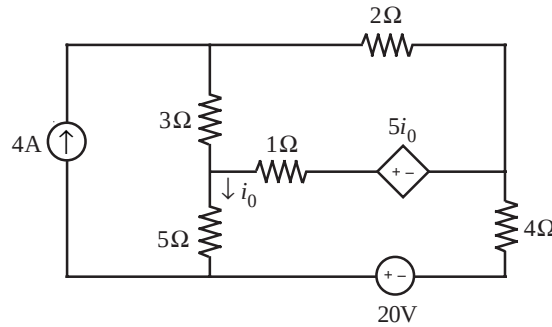


EC-3003-CBGS

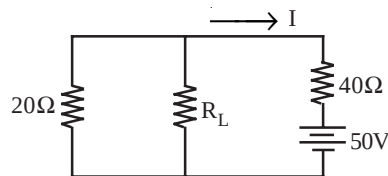
PTO

[2]

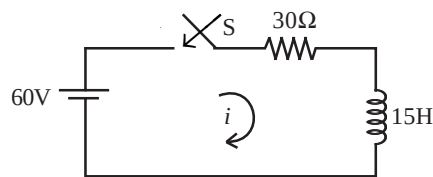
2. a) Find i_0 in the circuit shown using superposition. 7



- b) In the network of figure, the maximum power is delivered to R_L if its value is 7



3. a) A series RL circuit with $R = 30\Omega$ and $L = 15H$ has a constant voltage $V = 60V$ applied at $t = 0$ as shown in figure. Determine the current i , the voltage across resistor and the voltage across the inductor. 7



- b) Explain the transient analysis of series RC circuit. 7
4. a) Determine the Laplace transform of 7
- $e^{-at} \cos \omega_0 t u(t)$
 - $t^n e^{-at} u(t)$

EC-3003-CBGS

Contd....

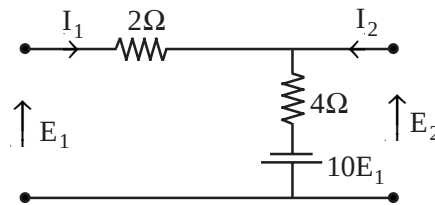
[3]

b) For the $X(s) = \frac{10(2s+3)}{s(s^2+2s+5)}$

Obtain initial value and final value.

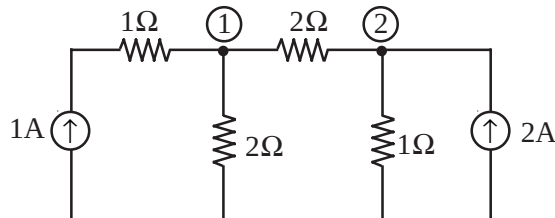
7

5. a) The z parameters z_{11} and z_{21} for the 2-port networks in figure are: 7

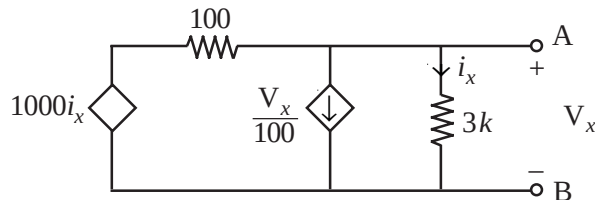


- b) Derive hybrid parameters in terms of z-parameters. 7

6. a) Find voltage at nodes 1 and 2 from the given figure: 7



- b) Find the Thevenin and Norton equivalent of the circuit. 7



[4]

- 7. a) Explain initial and final value theorem. 7
- b) Write down the condition for Reciprocity and condition for symmetry. 7

- 8. Write short notes on: 14
- a) Incidence matrix and tie set matrices
- b) Millman's Theorem
- c) Dot convention and coupling coefficient.
